

Unit IV Medium Access Control Protocols

explain mac protocol in detail

• defn of MAC

MAC - medium access control (MAC) is a sublayer of data link layer that controls/manages how multiple devices share a common communication channel without colliding / collisions

MAC is a sublayer of network layer that controls the transmission of data over a common communication channel without collisions.

• key concept of MAC

① Purpose of MAC: To avoid collision between devices

② Location: MAC is a sublayer of data link layer

③ Key Function: Decide which device can transmit data at a particular time

• Detect & handle collisions

④ MAC Address: MAC address is a unique

physical address used to identify a device in a network

• 6 bytes (48 bits) long

Types

① Random Access Protocol

② Controlled Access Protocol

③ Channelization Protocol

① Aloha

② CSMA

③ CSMA/CD

④ CSMA/CA

Reservation

Polling

Token passing

TDMA FDMA CDMA

Types of Medium Access Control (MAC) Protocols.

MAC

① Random Access Protocols

② Controlled Access Protocols

③ Channelization Protocols

① Aloha

② CSMA

③ CSMA/CD

④ CSMA/CA

① Reservation

② Polling

③ Token passing

① FDM

② TDMA

③ CDMA

① Random Access Protocols

Random Access Protocols is a type of MAC (Medium Access Control) in which all stations (any device) have equal rights to access communication channel at any time without waiting for permission.

or

Random Access Protocols is a MAC protocol where any station can use the channel at any time without waiting for permission.

Types of Random Access Protocols.

① Aloha

② CSMA

③ CSMA/CD

④ CSMA/CA

Pure Aloha

Slotted Aloha

Reservation

Polling

Token passing

① Aloha

② CSMA

Random Access Protocols is a type of MAC (Medium Access Control) in which all stations (any device) have equal rights to access communication channel at any time without waiting for permission.

1. ALOHA Protocol

ALOHA is the earliest random access protocol developed at the University of Hawaii.

In this protocol a station transmits data whenever it has data to send without checking whether the channel is free or busy.

Working

1. Station sends a frame immediately after sending frame.
2. If no collision occurs, transmission is successful.
3. If a collision occurs, the station waits for a

Working of Pure ALOHA

1. Station sends frame at any time.
2. After sending frame the station waits for ACK from receiver.
3. If ACK is received within specific time no collision occurred data is transmitted successfully.
4. If ACK is not received within specific time, station assumes collision occurred and waits for random amount of time to retransmit the frame (resend frame).

Advantage

- Simple & easy to implement.
- Suitable for small networks.

Disadvantage

- high chance of collision.
- Low efficiency $\rightarrow 18.4\%$.
- Poor performance under heavy traffic.
- Low throughput $\rightarrow$ means no of successful frames received.

Types of ALOHA

- $\rightarrow$ 1. Pure ALOHA
- $\rightarrow$ 2. Slotted ALOHA

1. Pure ALOHA (same as ALOHA)

In pure ALOHA a station transmits data any time whenever it has a frame to send.

Working

- Station sends frame at any time.
- If frame is after sending from the station

ALOHA = PURE ALOHA

maximum efficiency $\rightarrow 18.4\%$

2. Slotted ALOHA

In slotted ALOHA, time is divided into equal slots and each station (device) is allowed to transmit only at the beginning of a time slot. It reduces collision compared to pure ALOHA.

Working

The time is divided into equal slots (fixed no of slots). The station transmits the frame only at the beginning of a time slot.

- If the frame is received successfully an ACK is sent by the receiver.
- If ACK is not received within specific time station assumes that collision occurred.

After collision station waits for random number of slots before transmitting another frame.

- This is done to avoid repeated collision.
- Then after random number of slots, it retransmit frame.

Characteristics

- ① Synchronization required among stations
- ② Reduce collision compared to Pure Aloha

Advantage

- Less collision compare to Pure Aloha
- maximum efficiency = 36.8%
- more throughput compare to Pure Aloha

Disadvantage

- collision still occur
- more time required compared to Pure Aloha
- more complex compare to Pure Aloha.

Pure Aloha

① Definition: In pure Aloha station

transmit data any time whenever it has data (frame) to send

Slotted Aloha

① In slotted Aloha time

is divided into equal slots and each station allowed to transmit only at beginning of time slot

② Synchronization required

No synchronization

More collision

③ Less collision

18.4%

④ 36.8%

Simple implementation

⑤ complex implementation

Lower throughput

⑥ More throughput

Throughput means no. of successful frame received.

① ~~more~~ more retransmissions are needed

④ ~~less~~ fewer retransmissions are needed

② poor channel utilization

③ better channel utilization

⑦ suitable for every small network

⑧ suitable for networking with higher network

⑨ older & basic version of Aloha

⑥ improved version of Aloha

Aloha

type

types of Aloha

① Pure Aloha → pure

② Slotted Aloha → working

come up points

characteristics

disadvantages

efficiency

Explain CSMA & its working principle classmate

II CSMA (Carrier sense Multiple Access)

defn CSMA: CSMA is a medium access ^{control} protocol in which a station first listens to the channel before transmitting data.

- If the channel is free it transmits data, if the channel is busy it waits until the channel becomes free.
- It works on principle of 'Listen Before Talking' or 'sense before transmit'.

Types of CSMA Access mode

- ① 1-Persistent CSMA ② Non-Persistent CSMA
③ P-Persistent CSMA

① 1-Persistent CSMA

- In 1-persistent CSMA station senses / listens channel continuously.

• If the channel is ^{free} idle it transmits data.

defn immediately if it is not idle it does not transmit data. ^{station} continuously senses channel until the channel becomes free.

② Non-persistent CSMA

A CSMA access mode in which station waits for a random period of time before sensing again.

③ P-Persistent CSMA

In P-persistent ~~time~~ time is divided into different slots, if the channel is idle, station transmits data with probability p and wait for next slot with probability $(1-p)$.

Random backoff period $\rightarrow$ random waiting period after collision

Working of CSMA

1. station just sense (listen to) the channel
2. If channel is idle (free), the station transmits the data
3. If the channel is busy, the station waits according to CSMA access mode such as 1-persistent, non-persistent, p-persistent
4. After channel becomes free the station retransmit data
5. If 2 stations transmit simultaneously a collision may occur
6. After collision, the stations wait for a random time and retransmit the data

Explain CSMA/CD, CSMA/CA

Types of CSMA Protocol

- 1. CSMA/CD (Collision detection)
- 2. CSMA/CA (Collision avoidance)

1. CSMA/CD (Carrier sense multiple access with collision detection)

CSMA/CD is a protocol in which station senses the channel before transmission & detects collisions during transmission

How it works? (working)

1. If the channel is idle, transmit data
2. If it detects a collision, it stops & keeps the transmission immediately
3. Send a jam signal and wait for a random backoff time before retrying
4. Jam signal is a burst to every one (collision)

Retransmit the data.

Usage: - Used in Ethernet (IEEE 802.3)

by concept of CSMA/CD

1. Channel checking/sensing - check if channel is free before sending
2. Collision Detection - detect whether 1 & signals collide while transmitting
3. Collision handling $\rightarrow$ Once a collision is detected station sends 'jam signal' to notify others about collision, it stops transmission & wait for random time before retransmission.

Detecting collision after

try error

2 CSMA/CA (Carrier sense Multiple Access with Collision Avoidance)

CSMA/CA is a type of CSMA protocol ~~that~~ in which station checks the channel before sending data & tries to avoid collision before sending data & tries to avoid collision by waiting & using ACK

Working of CSMA/CA

- 1 station senses the channel whether it is free
- 2 if the channel is busy, it waits for random back off time
- 3 if the channel is free, it waits for a short time & then sends data back off time
- 4 to avoid collision, the station do not transmit data immediately they wait before sending the data (avoidance of collision before station sends data)
- 5 after sending data the sender wait for ACK from receiver
- 6 if ACK is received, transmission is successful.
- 7 if ACK not received it assumes retransmit data after some time

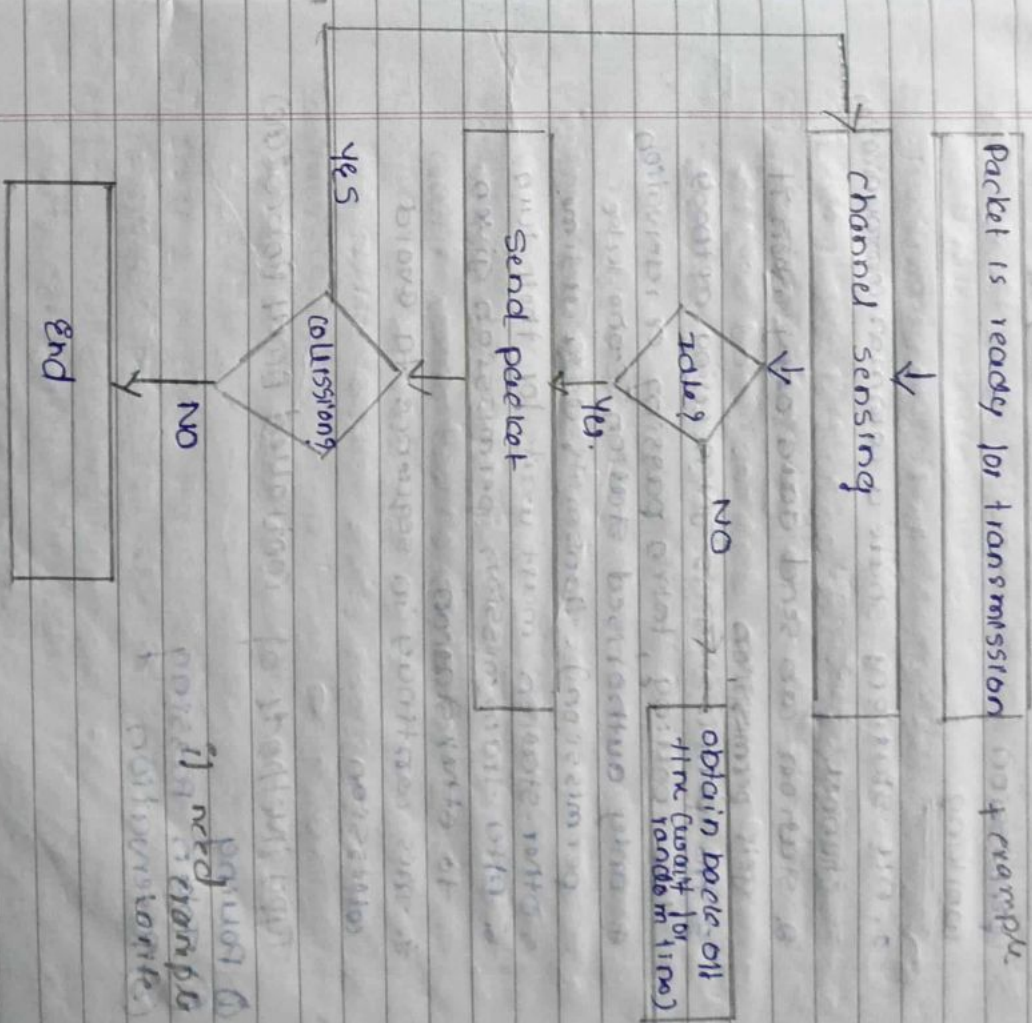
Usage: used in wireless networks like wifi

Key concept/aspect of CSMA/CA

- 1 channel checking :- check whether the channel is free or not.
- 2 collision avoidance :- if the channel is busy the station waits for a random period of time before retransmitting reduce chance of collision

- 3 ACK :- After successful transmission the receiver sends an ACK to confirm ensuring data correctly

Diagram



Controlled Access Protocol (CAP)

defn: Controlled Access Protocol is a method in which transmission is allowed only when permission is given, so that only one station transmits at a time to avoid collision.

permission working

1. All stations share a common communication channel.
2. Station can send data only when it gets permission.
3. The permission is given using methods like polling, token passing or reservation.
4. Only authorised station can take permission to transmit data at time.
5. Other station must wait for their turn.
6. After transmission permission given to other station.
7. This continues in sequence to avoid collision.

Types/Methods of Controlled Access Protocol (CAP)

1. Polling
2. Token Passing
3. Reservation

Polling

defn: Polling is a method of controlled access protocol in which a central controller (master) checks each station one by one and gives permission to transmit data.

Working

1. A central device called as master controls all communication.
2. The master sends poll message to each station in order.
3. Poll message is a message which is sent by master to a station to ask whether it has data to send.
4. The station which has data and chosen to send data.
5. If the station has data, it transmits it to the receiver.
6. If the station has no data, it sends 'no data' response.
7. The master then moves to the next station in sequence.
8. This continues in a round-robin manner.

Key points

1. Only 1 station transmits data at time.
2. Controlled by central master.
3. No collision occurs.
4. Each station gets equal chance.

Advantage

- No collision
- Easy to manage
- Equal opportunity to all station

Disadvantage

- If master fails system stops
- slow if many stations exist
- wastage of time if station they not have data

Token Passing

Token token passing is a method of controlled access in which a special token is passed between stations, and only the station with the token can send data

Working

1. A special frame called token is created in network
2. The token is passed from one station to another station in a fixed order either a ring or logical sequence
3. Each stations check tokens when it receives it
 - If means no station wanted it to transmit data or just pass the token to next station
4. If the station has no data it simply pass the token to next station
5. If it has data it captures token and sends data (transmits data)
6. After completing transmission of data the station releases token.

∴ the token then move to the next station and this process continues in a cycle.

Key points

1. Only the station with token can transmit data
2. token circulate in fixed order
3. No collision
4. Equal opportunity to all stations

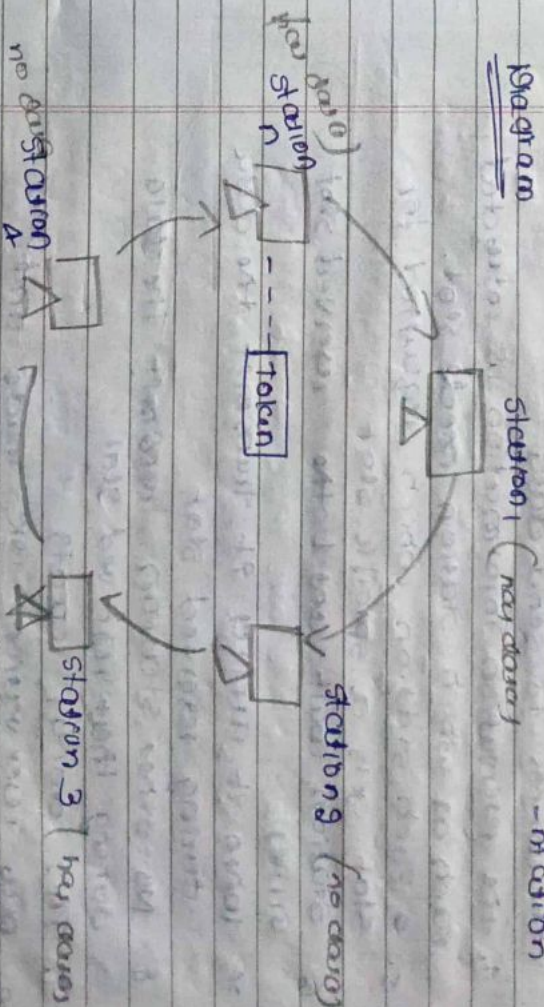
Advantage

- No collision
- Efficient in heavy traffic
- Predictable transmission time

Disadvantage

- Token can stop network
- Time consuming
- Complex implementation

Diagram



For ex:- If there are 4 stations, a special packet

called as token circulate among all station first it reaches S1 if it has data, S1 transmits data capturing token and transmits data after transmission is over, token pass to next station S2 if it has no data

Reservation

Q1 Reservation is a controlled access method in which stations reserve time slots before transmission so that only one station can send data in a given period of time.

Working.

1. The channel ^{time} is divided into different slots.
2. Before transmitting data, each station ^{each} reserve a slot. It gets chance to receive data.
3. Station informs the network that it wants to send data.
4. The reservation information is collected such as which station needs slot.
5. Each station which requested for slot gets a specific slot.
6. Station waits until the reserved slot arrives.
7. When it arrives it transmits the data during reserved slot.
8. The other station transmits the data during their reserved slot.
9. This process repeats.
10. Now reservation is made and transmission continues.

Ex 1 1) A, B and C share communication channel.

- 1) A & C want to send data.
- A reserves slot 1.
- C reserves slot 2.
- A sends data in slot 1.
- C sends data in slot 2.
- No collision occurs.

Key points

- Time divided into slots.
- Stations must reserve before sending.
- Only one station transmits per slot.
- No collision.
- Organised transmission.

Advantage

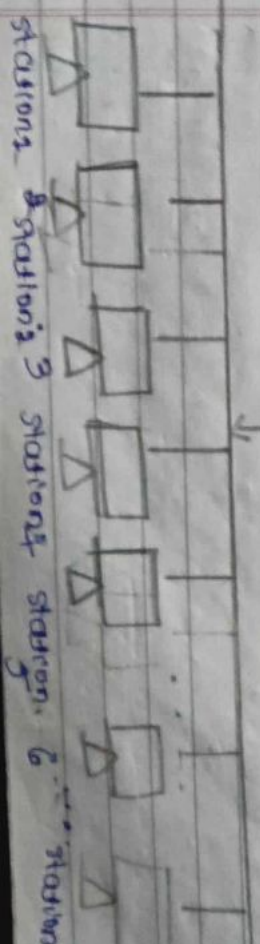
- 1) No collision.
- 2) Scheduled transmission.
- 3) Equal chance to all stations.

Disadvantage

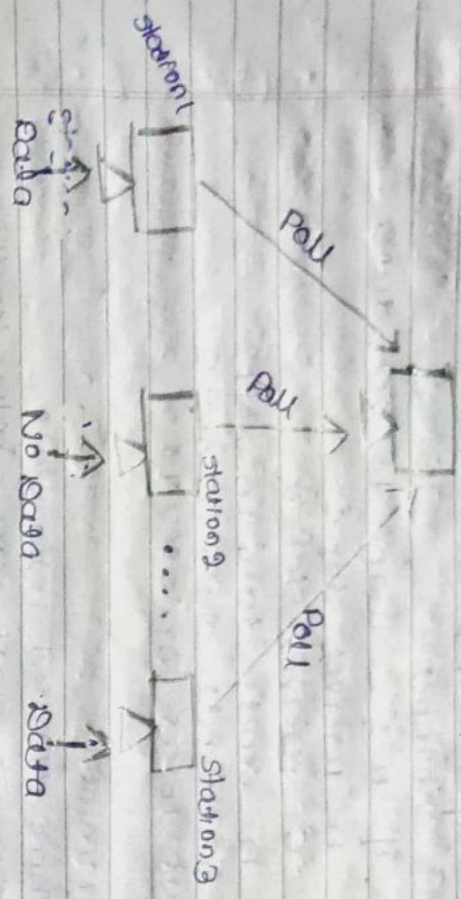
- 1) Delay due to reservation process.
- 2) Complex system.
- 3) Wastage of slots if station has no data.
- 4) Not good for small traffic.

Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6	Slot 7
0	1	0	1	0	1	0

1 - station which has data
0 - station has no data



Polling diagram central controller/master



Example Consider 3 stations Station 1, Station 2, Station 3

(i) Controller/master send ~~mess~~ poll message to station 1

Station 1 has data S1 transmits data.

(ii) master sends poll message to S2, it does not have data move to next station

(iii) master sends poll to S3, it has data so S3 transmit data

So thus it avoids collision

3 type in MAC - Channelization → Shared Difference CSMA/CD & CSMA/CA

CSMA/CD

CSMA/CA

- | | |
|---|--|
| <p>standby for carrier sense multiple access with collision detection</p> <p>CSMA/CD is a type of CSMA protocol in which station sense channel before transmission & detect collision during transmission</p> <p>③ detect collision after they occur</p> <p>④ used in wired ethernet network</p> <p>⑤ station transmit the data immediately when channel is free</p> <p>⑥ use collision detection mechanism</p> <p>⑦ Send 'Jam signal' when collision occurs</p> <p>⑧ retransmit data after random backoff time.</p> <p>⑨ more suitable for wired media</p> <p>⑩ simpler protocol for collision detection</p> | <p>standby for carrier sense multiple access with collision avoidance</p> <p>③ CSMA/CA is a type of CSMA protocol in which station checks the channel before sending data & try to avoid collision before sending data</p> <p>⑤ try to avoid collision before they occur</p> <p>⑥ used in wireless network.</p> <p>⑦ station waits & use avoidance technique before transmitting</p> <p>⑧ use collision avoidance mechanism</p> <p>⑨ uses RTS/CTS request to send (clear to send)</p> <p>⑩ wait for ACK & use backoff to avoid collision</p> <p>⑪ collision media.</p> <p>⑫ more complex protocol</p> <p>⑬ it minimize collision as much as possible before transmission</p> |
|---|--|

Ethernet and IEEE 802.3

IEEE → Institute of electrical and electronics engineers

Ethernet is a set of technology and protocols that is primarily used in LAN's

~~But~~ It was standardised by IEEE Q9

IEEE 802.3 9m 19805

at defining physical layer and the sub-layer of Data Link layer that is MAC (Medium Access Control)

- It is used for communication in wired LAN category.

Pharm (Lee 808.3) Frame Form

packet	S F D	destination address	source address	length	checksum	data	crc
7 bytes	1 bytes	6 bytes	6 bytes	2 bytes	2 bytes	1500 bytes	4 bytes

① Preamble,

① Preamble is a first yield of document from

③ 7 + 15 = 22 bytes payload.

③ It contains alternating 0's and 1's

④ pattern is 0101010101...

(5) The preamble helps receiver to adjust timing with sender so that data can be exactly received correctly.

helps to achieve bft synchronization)

② SFD (Start of Frame)

① SFD comes after Preamble.

⑧ This is a byte field

③ It is actual beginning of Ethernet frame

④ $\text{pH} = 7.4$ pattern is 10101011

⑤ next p100 q11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845,

③ Destination Address

① It comes after SFD

② This is a dog print

④ The pyrame is delivered to trans destination address!

④ Source Address

① Source Reliability comes with destination accuracy

⑥ This is 6 byte field

③ This contains MAC address of the sender.

④ source address tells who send the data

⑤ length

⑥ This link comes after source address

② This is a byte field

③ It indicates the length of the entire chain or name.

④ length can not be larger than 1500 bytes.

⑤ maximum length 1500 bytes

6) minimum length 46 bytes.

- ① Data.
 - ① this field comes after length
 - ② This field contains actual information being transmitted
 - ③ It is also known as Payload
 - ④ contains IP header and data.
 - ⑤ minimum size is 46 byte
 - ⑥ maximum size is 1500 byte
 - ⑦ most important part of frame

- ① CRC - Cyclic Redundancy Check (CRC)
 - ① CRC comes after data field
 - ② This is 4 byte field
 - ③ This is used for error detection
 - ④ check whether data is received correctly
 - ⑤ Detects damaged or corrupted data
 - ⑥ Last field of Ethernet frame

Types of Ethernet IEEE 802.3

- ① Standard Ethernet - speed 10 Mbps
used for basic network communication
ex! - 10BaseT
- ② Fast Ethernet - speed 100 Mbps
Faster than standard Ethernet
ex! - 100 BaseTx
- ③ Gigabit → speed - 1000 Mbps
provides high speed data transfer
ex! - 1000 Base-T
- ④ 10 Gigabit Ethernet → speed 10 Gbps
used in high performance networks
ex! - 10GBase-T

10M

HDLCL

Bit Rate Based rate.

classmate

Date _____
Page _____

High Level Data Link Control (HDLC)

defn HDLC is a bit-oriented Data Link Layer protocol used for used as the Data Link layer to send data accurately and safely between devices.

- Bit oriented protocol.
- Provides error control
- support flow control.
- used in point to point & multipoint
- works at data link layer,

HDLC Frame Format

Beginning Sequence (Flag)	Header	Body	CRC	ending Sequence
8 bits	16 bits		16 bits	8 bits

FIELD of HDLC frame

- (1) Beginning sequence / Flag field ^{start of frame}
 - Indicates the beginning & end of a frame
 - It is a 8 bits field
 - Has a fixed pattern 01111110
 - It helps the synchronization b/w sender and receiver
 - The Beginning Sequence allows the receiver to adjust its timing with the sender so that data received correctly

② Header

- Header contains control information needed for communication
- Header field contains Address field and control field

① Address Field

- contains the address of sender or receiver.
- Helps to identify the destination station.
- If the frame is sent by primary station, the address field contains destination - "To" address of secondary station
- If the frame is sent by secondary station, the address field contains the source - "From" address of the secondary station

(note)

② Control Field

- control field is used for flow & error control
- controls data transmission

③ Body (Payload Field)

- contain actual user data.
- This field contains user data & other information
- It is the main part of frame
- length of data can vary

④ CRC / FCS (Frame check sequence)

- This is error detection field
- If error is found, the frame can be retransmitted.
- checks the data received correctly
- CRC stands for Cyclic redundancy check

⑤ Ending Sequence

- Indicates end of ~~data~~ frame
- Helps the receiver to recognize where the frame ends.
- Same pattern as beginning : 01111110

Types of HDLC Frame

There are 3 types of HDLC Frame

① Information Frame (I-Frame)

def / This frame is used to transport actual user data b/w sender & receiver.

- used for normal data transmission
- supports error & flow control.

② Supervisory Frame (S-Frame)

def / used to control & monitor data transmission

- used for flow control
- used for error control.
- Does not carry user data
- sends ACK

③ Unnumbered Frame (U-Frame)

def / Used for system management or link management

- used to establish or terminate connection
- Do not carry normal user data.

Control field determining the type of HDLC Frame

I-Frame	1st bit is 0
S-Frame	1st two bits is 10
U-Frame	1st two bits is 11

- ① If first bit is 0 of control field identifies information frame
- ② If first 2 bits are 10 then it is Supervisory frame
- ③ If first 2 bits is 11 then it is U-Frame

Disadvantages of HDLC

- More complex than simple protocols
- Requires additional processing
- Not commonly used in modern LAN

Application of HDLC

- ① WAN
- ② Point-to-Point communication
- ③ Satellite communication
- ④ Data transmission systems

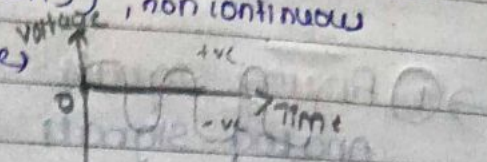
II Digital Transmission & Physical Layer

Digital Representation of Data

Data representation

Analog - continuous, both integer & decimal (voice, temp, weather, clock)
 Digital - Discrete - 0 & 1 (bits), non continuous

signals $\rightarrow$ represented in data wave



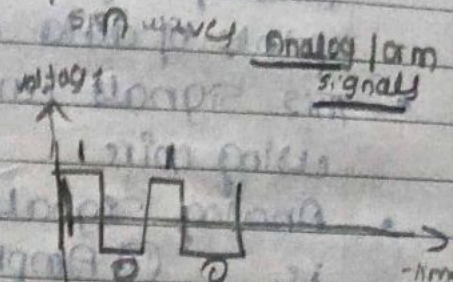
Analog

we get specific points in wave form

OST reference model

- 7 application
- 6 presentation
- 5 session
- 4 transport
- 3 Network

- 2 data link
- 1 physical layer

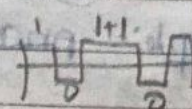


Digital signals

Digital representation of data

Signals

- 3 parameters of signal characteristics:
- ① Amplitude
 - ② frequency
 - ③ phase



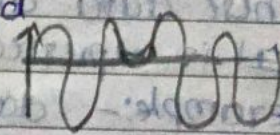
101101

① Amplitude $\rightarrow$ height of wave / displacement of wave

② frequency $\rightarrow$ no. of cycle / oscillation per second

③ phase $\rightarrow$ position of wave

(either analog repn or digital repn)



no. of cycle passed in 1 second

Digital Representation of Data
Data and signals can be represented either in analog or digital form

- ① Analog signals refers to continuous set of specific points of data and all the possible points in between. This signals can be transmitted using air or using wire

Analog signals are described in 3 parameters
① Amplitude
② Frequency
③ Phase

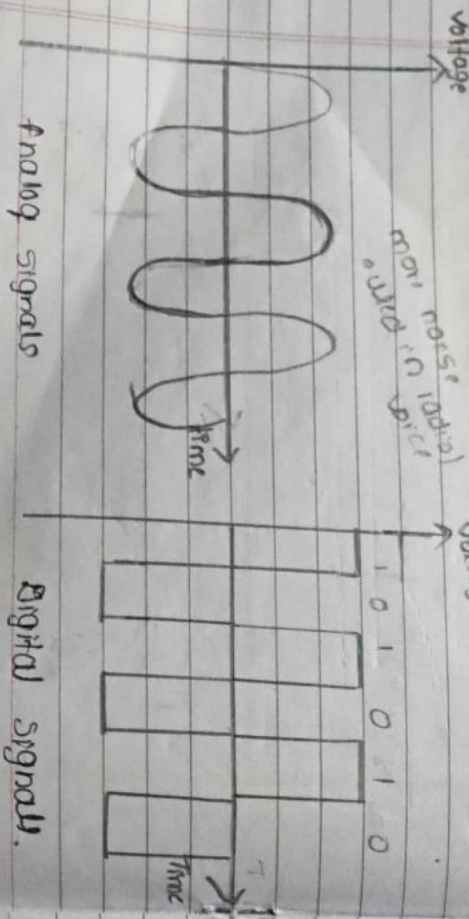
example:- analog clock, radio waves, human voice

- ② Digital signals refers to the discrete set of specific points with no data points in between.

These data are stored in computer memory

It is captured in bits per second
example:- data stored in CD, DVD and floppy disc.

data digital watch/clock

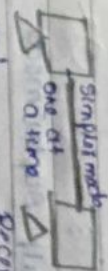


Digital Transmission - Transmission modes

Transmitting data via a device or having network is called transmission mode. It is divided into two types:

1. Simplex
2. Half-Duplex
3. Full-Duplex

① Simplex mode



- unidirectional
- sender is 1

sender Transmitter

receiver

• data is sending one direction

Digital transmission

Transmission mode or communication mode.

Transmission mode refers to the transferring of data plus deciding how data is transferred.

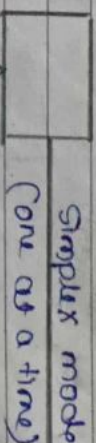
- It is also known as communication mode.
- There are 3 types of Transmission mode.

- ① Simplex mode
② Half-Duplex mode
③ Full-Duplex mode

- ① Simplex Mode
In simplex mode the communication is unidirectional.

- ② In this sender can send the data but the sender can not receive the data.

example:- keyboard, mouse, monitor etc communication



Simplex mode (one at a time)

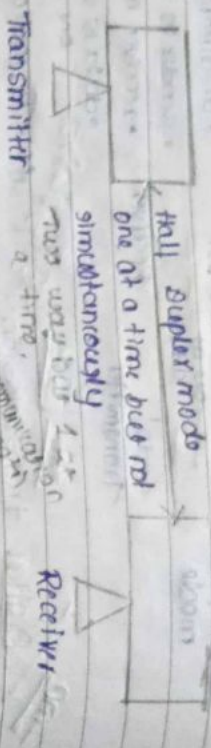
Transmitter

Receiver

② Half Duplex mode

In a half duplex mode the communication is in a half duplex mode but not simultaneously. In this each can transmit and receive the data but not at the same time. This is used in the case where there is a need of communication in the both direction at the same time.

Ex:- walkie-talkie.



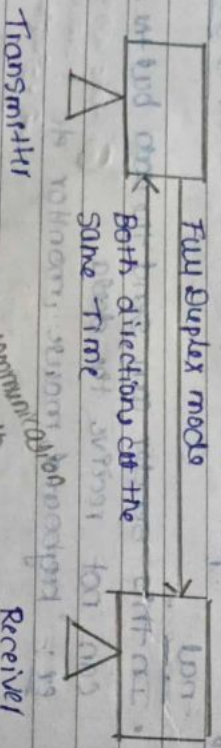
③ Full Duplex Mode

In full duplex mode the communication is bi-directional and simultaneously.

In this both stations can transmit and receive the data simultaneously.

In this mode the line must contain 2 separate transmission paths one for sending & other for receiving.

Ex:- Mobile, landline, telephone calls.



② Modulation and Demodulation
Modulation is a process of changing the characteristics of the carrier signal (Amplitude, Frequency, Phase) with respect to the message signal.

③ Types of Modulation are
① Analog Modulation
② Digital Modulation

① Analog Modulation
Analog modulation is a technique which is used in analog data signal transmission and digital signals.

Ex:- broadband signals of carrier signal in analog modulation
i.e. Amplitude Frequency and Phase

① Analog Modulation

Analog modulation is a technique which is used in analog data signal transmission and digital signals.

Ex:- broadband signals of carrier signal in analog modulation
i.e. Amplitude Frequency and Phase

The three types of analog modulation are
1. Amplitude modulation (AM)
2. Frequency modulation (FM)
3. Phase modulation (PM)

1. Amplitude Modulation
It is process of changing the amplitude of the carrier signal with the respect to the message signal by keeping frequency and phase constant.

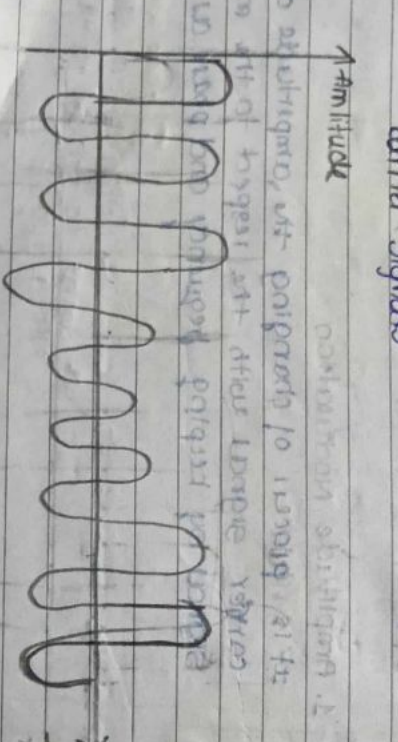
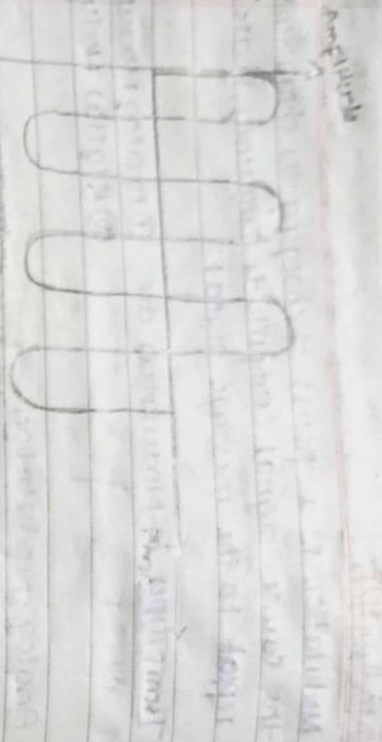
Message signal $\rightarrow$ original information signal
we want to send
of low frequency
carrier signal $\rightarrow$ high frequency signal used to carry message signal to long distance.

Message signal $\rightarrow$ original information signal
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Message signal $\rightarrow$ original information signal
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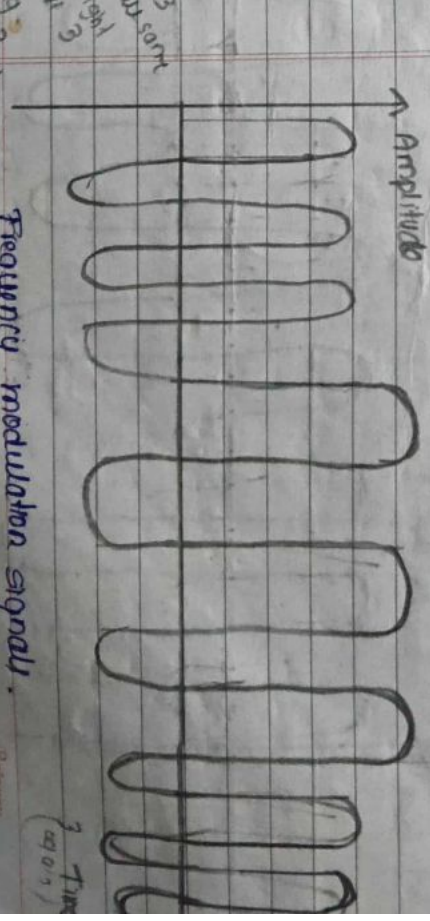
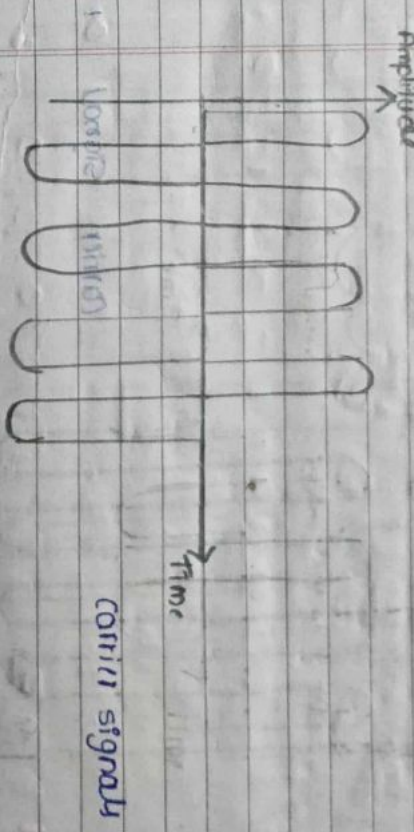
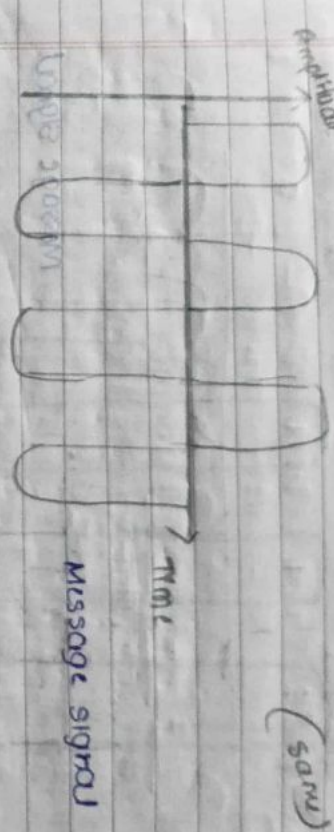
Message signal $\rightarrow$ original information signal
we want to send
of low frequency
carrier signal $\rightarrow$ high frequency signal used to carry message signal to long distance.

Message signal $\rightarrow$ original information signal
we want to send
of low frequency
carrier signal $\rightarrow$ high frequency signal used to carry message signal to long distance.



Amplitude modulation signal

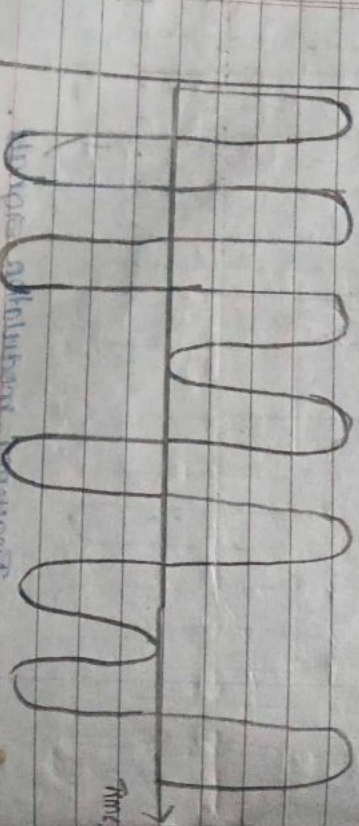
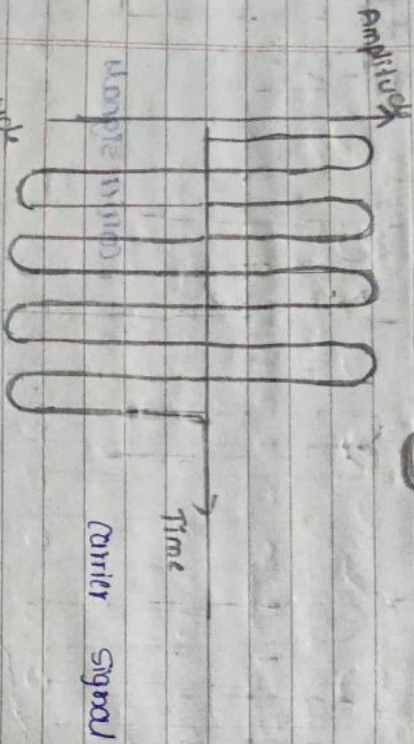
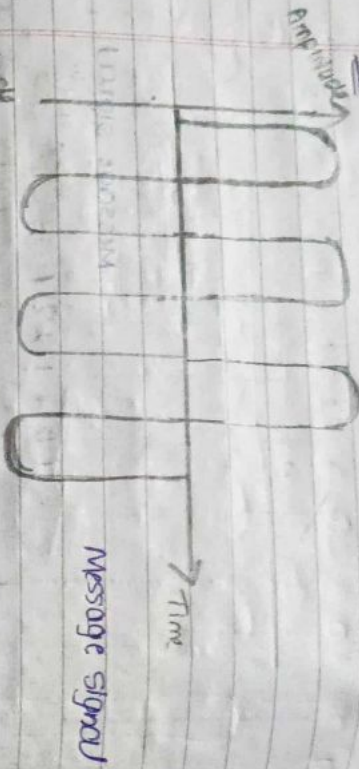
2. Frequency Modulation
It is a process of changing the frequency of the carrier signal with respect to the message signal keeping amplitude and phase constant.
ex- FM radio, broadcasting.



Frequency modulation signal

3. Phase modulation

It is a process of changing the phase of carrier signal with input message signal keeping amplitude and frequency constant. eg: wi-fi, satellite television.



1. Digital Modulation

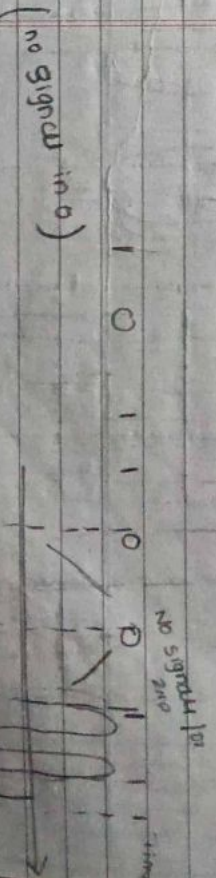
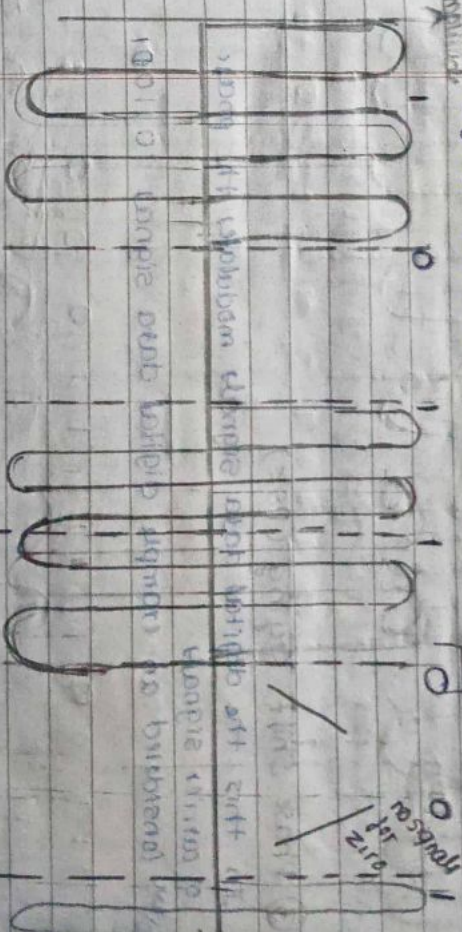
Digital modulation is a process of converting digital or discrete data signals into analog data signals.

There are 3 types of digital modulation, they are:

1. Amplitude Shift Keying (ASK)
2. Frequency Shift Keying (FSK)
3. Phase Shift Keying (PSK)

1. Amplitude Shift Keying (ASK)
In this the digital data signals modulate the amplitude of carrier signal.

eg: digital data = 1 0 1 1 0 0 1



Phase modulation signal

frequency ↑
wavelength ↓
increased frequency

⑧ Frequency shift keying (FSK)
In this the digital data signals modulate the frequency of carrier signals

Considered an example of digital data signals 1011001



③ Phase Shift Keying (PSK)

In this the digital data signals modulate the phase of carrier signals

Considered an example digital data signal 1011001



Line Coding / Line Encoding
is a coding is a process of converting digital data into digital signals.
At the sender the digital data are encoded into digital signal.

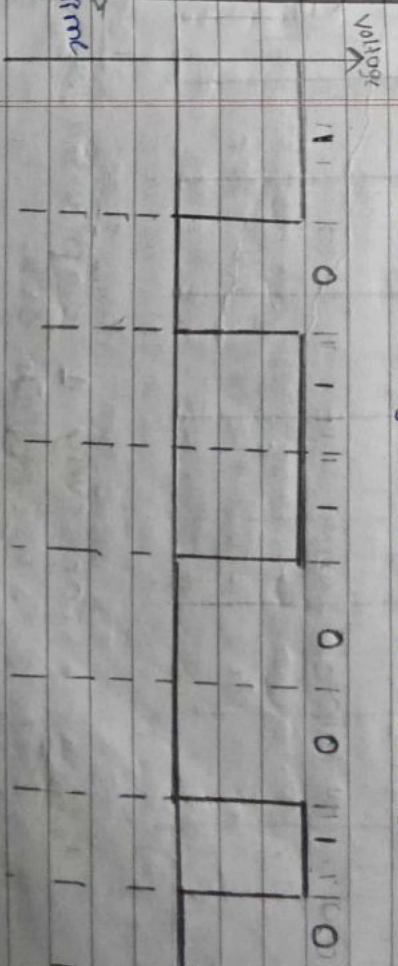
There are 3 types of line encoding schemes
1. Unipolar
2. Polar
3. Bipolar

Unipolar coding schemes convert 1 to +V and 0 to 0V. In this scheme all the signal levels are either above or below the GND.

In unipolar line coding scheme the positive voltage defines bit 1 and the zero voltage defines bit 0.

- 1 → +ve
- 0 → zero voltage

Considered the digital data 10110010



3. Polar In this scheme a signal is represented by a positive voltage and other by a negative voltage level.
i.e. on both the sides of axis

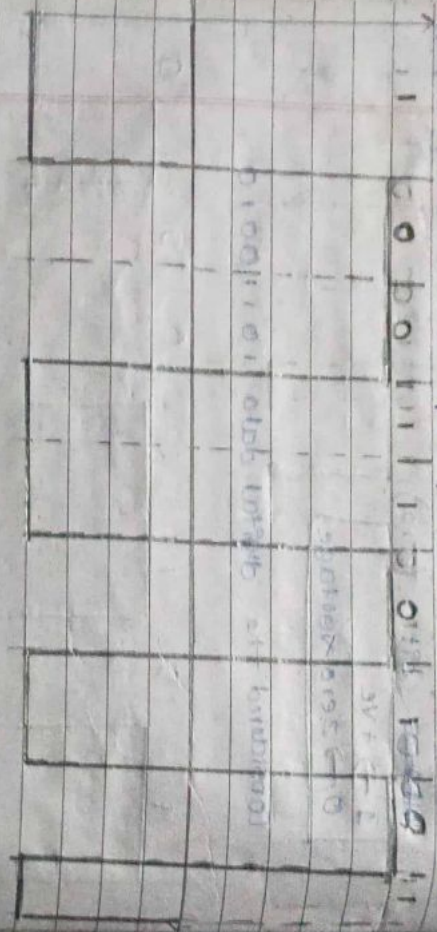
① Polar NRZ (Non-Return to Zero)
A few types of polar encoding schemes are

→ a) NRZ - L (Level)

In polar NRZ - L the voltage level determining the bit-1 ~~and~~ is represented by positive voltage and zero is represented by negative voltage

1 → +V
0 → -V

Consider a example of digital data 100110101



draw • For every 1 draw a straight line above axis.
• For every 0 draw a straight line below axis.

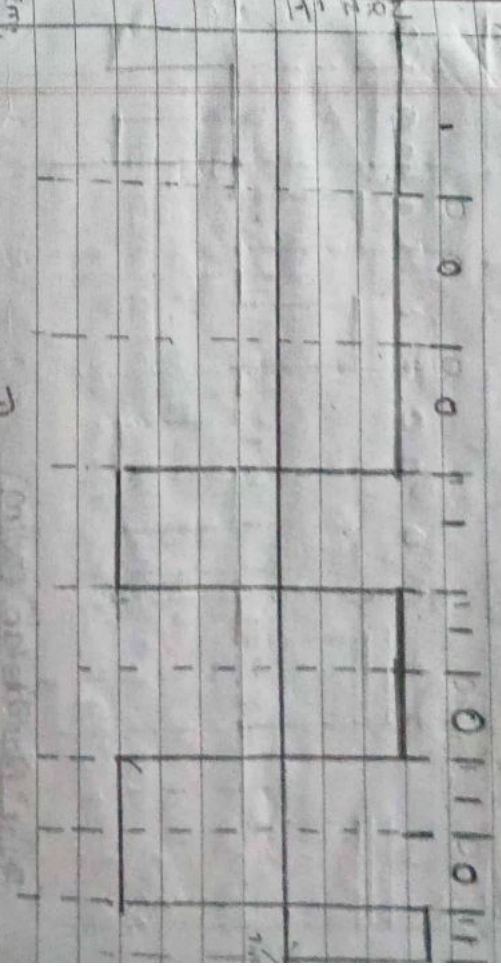
above axis

→ b) NRZ - I (Invert)

In polar NRZ - I transition or no transition in the voltage level determine the bit-0 logic. If it is bit-1 then a transition from one voltage level to another voltage level take place.
If it is bit-0 there is no transition of voltage level.

1 → Transition
0 → NO transition

Consider a example of digital data 100110101



(upward this)

Transition or not — 1-0

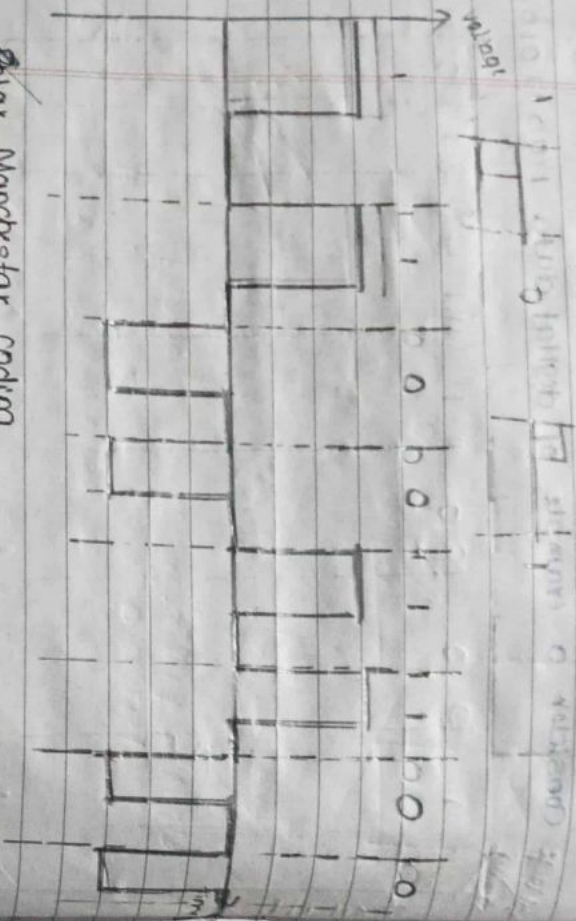
draw • If bit is 1 - change level

• If bit is 0 - continue same level

→ Polar RZ (Return to zero)

In polar RZ, If has a voltage level, i.e., positive, negative & zero voltage to encode the data. A transition from negative voltage to zero voltage level, indicating bit-0 and a transition from positive voltage to zero-voltage level indicates bit-1.

Consider a digital data 11001100



Polar Manchester Coding

In polar manchester coding, there is a transition of the middle of each bit a transition from negative to positive voltage level determines bit-1 and a transition from positive to negative voltage level determines bit-0.

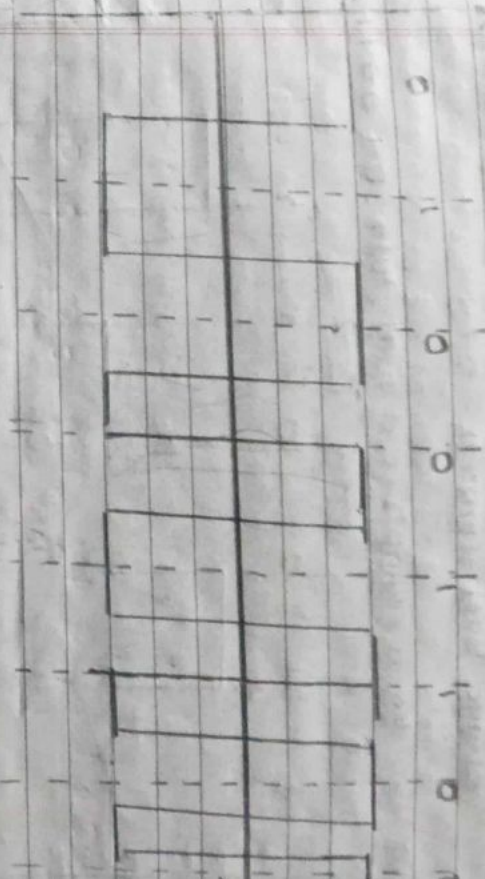
1 → -V to +V

0 → +V to -V

for 0 go down in the middle
for 1 go up in the middle

Draw: Divide each bit into 2 halves
Draw transition in the middle
for 0, go down in the middle
for 1, go up in the middle

Consider a example of digital data 01001100



Polar Differential Manchester Coding

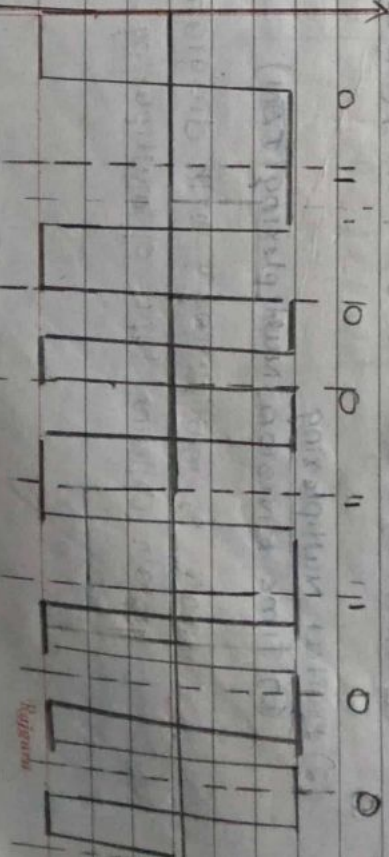
In polar differential manchester coding, the bit 0 is represented by a transition in the voltage level and bit-1 is represented by no transition in the voltage level.

Note that the transition always occurs from the negative voltage level.

0 - Transition

1 - No Transition

Consider a example of digital data 01001100



Ques. What is bipolar? Ans. Bipolar is a type of digital signal where the voltage level represents bit-0 and bit-1.

2. Bi Polar
In Bipolar a neutral or zero voltage level asking represents bit-0 and alternative positive and negative voltage level represents bit-1

Ques. Consider a example of digital data 0 1 0 0 1 1 1 0



4. IMP
Multiplexing is the technique of combining multiple signals into one signal so that they can be transmitted through single communication channel.
Multiplexing is a process of transferring multiple signals over a single communication channel.

IMP
Signal

The types of Multiplexing are

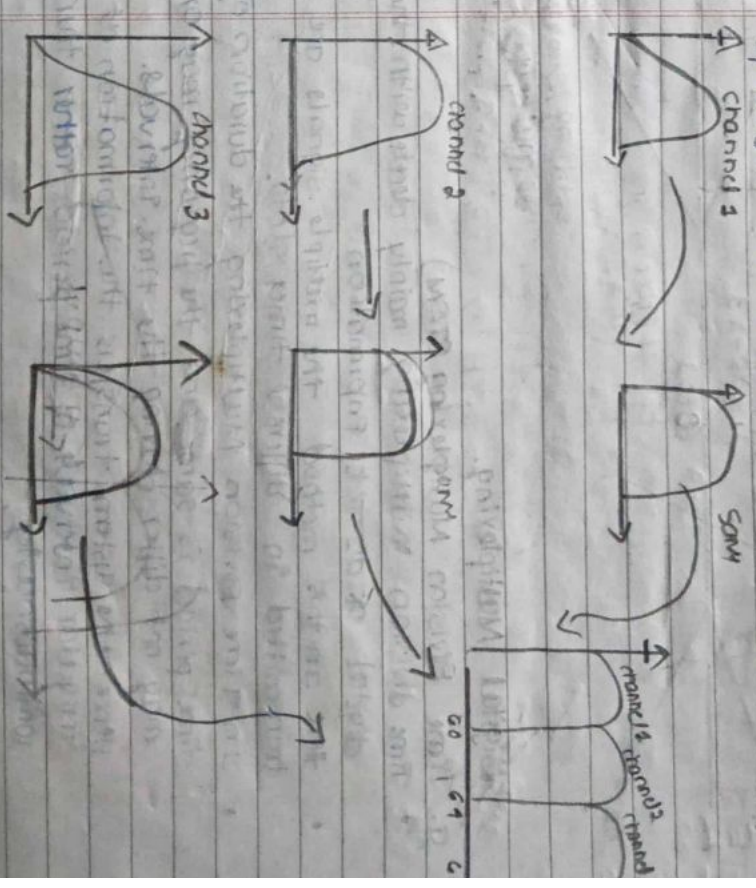
- ① Analog Multiplexing &
- ② Frequency Division Multiplexing (FDM)
- ③ Wave Length Division Multiplexing (WDM)
- ④ Digital Multiplexing
- ⑤ Time Division Multiplexing (TDM)

Ques. What is multiplexing? with diagram explain different type of multiplexing - 8m

Analog Multiplexing

① Frequency Division Multiplexing (FDM)
FDM is a analog technique which can be applied when the bandwidth of the communication channel is greater than combined bandwidth of the signals i.e. to be transmitted. In FDM signals generated by each sending device modulate signals by using different carrier signals. These modulated signals are combined into a single signal that can be transmitted over the communication channel.

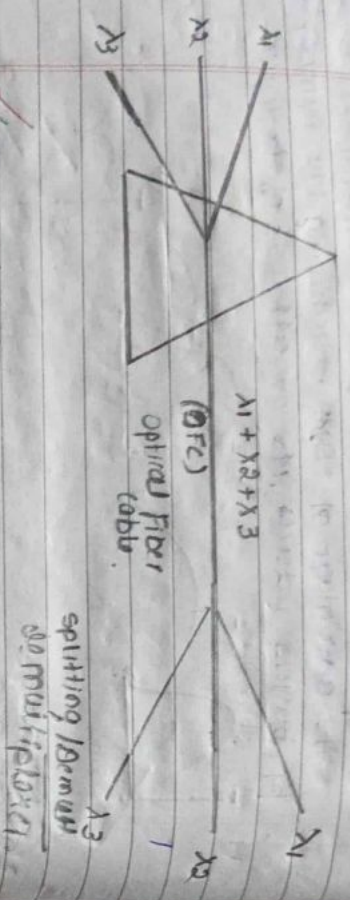
The advantage of FDM is reducing the number of buffers reduces the overall cost of the system.



Frequency

Wavelength Division Multiplexing (WDM)

In WDM, we use high frequency and combine different signals of different frequency. A light source from different sources are combined to make a band of light, at the receiver, the signals are separated by using demultiplexer, the combining and splitting and of light sources is handled by prism.



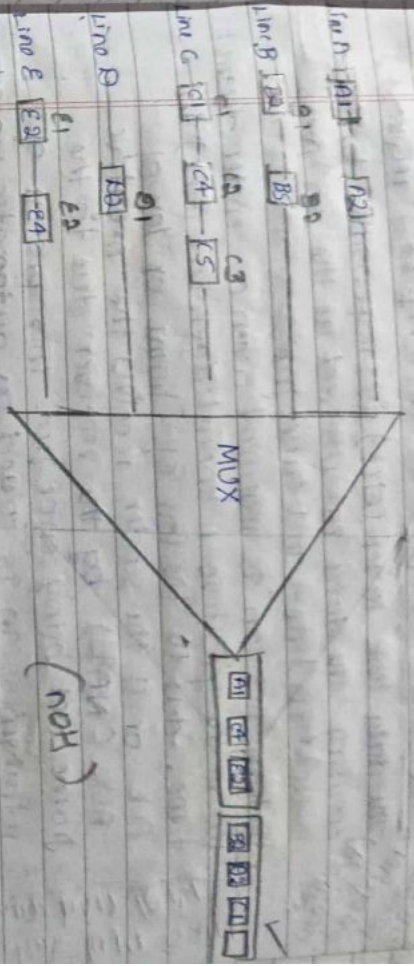
Digital Multiplexing

a. Time Division Multiplexing (TDM)

- Time division multiplexing mainly deals with the digital or discrete information.
 - Here in this method the multiple signals are transmitted in different time slots.
 - In Time Division Multiplexing the duration of time period is same and the frequency may or may not differ during the time intervals.
- Hence the system transmits the information at regular intervals of time period rather than continuously.

Input of TDM

Asynchronous / Multiserial Multiplexing
data + address



Types of TDM $\rightarrow$ Least priority

Data Link Layer (DLL)

Protocol's for automatic Repeat request

Ensuring the data is received at the receiver end in line and error free

Explain data link layer issues

Retransmission is required when a sender doesn't receive the positive Acknowledgement (ACK) in time, due to the loss of frame or loss of ACK or if the sender receives the negative ACK (NAK) by the receiver due to the frame not being error free. This process of retransmission is called as automatic repeat request (ARQ).

Then one commonly 3 methods for flow and error control they are

1. Stop & Wait ARQ

This is the simplest method for flow and error control. This protocol is based on the concept that the sender will send the frame and wait for the ACK until the ACK is received.

Then one 4 possible outcomes if the transmission of the frame or ACK leads to an error. They are

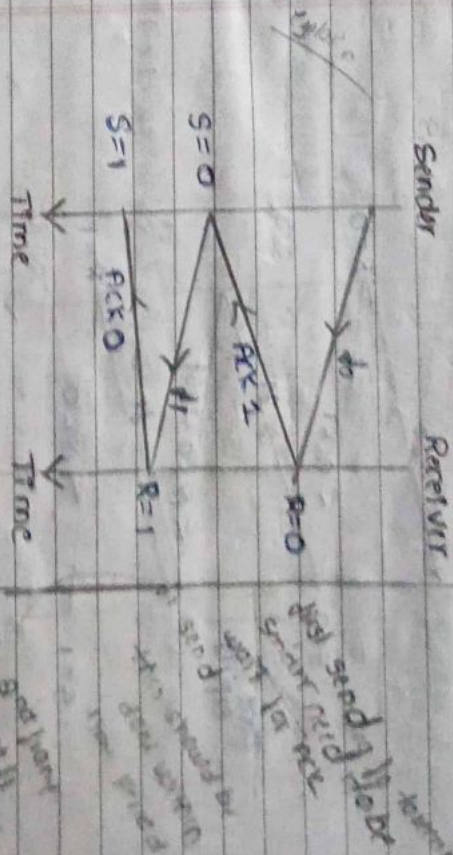
- Normal operation
- When ACK is lost
- When frame is lost
- When ACK time out occurs

Explain data link layer design issues

Normal Operation

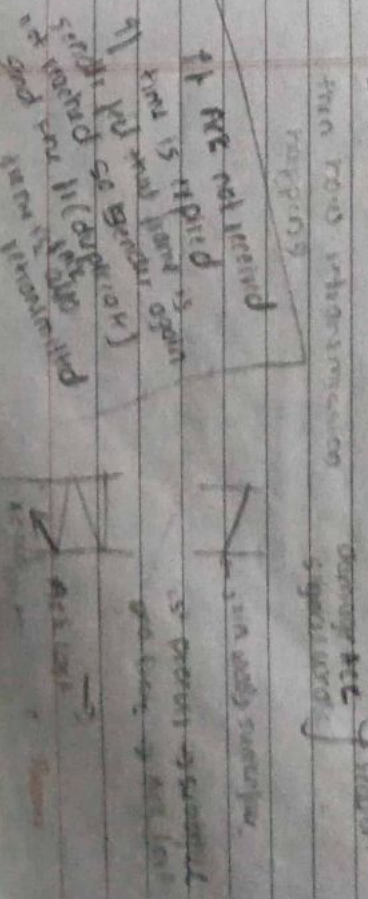
If the sender is sending frame 0 then it will wait for the ACK 1 which will be transmitted by the receiver with the expectation of the next frame numbered 1.

As it receives ACK 1 in the given allotted time this process will be repeated until the complete data transmission takes place.



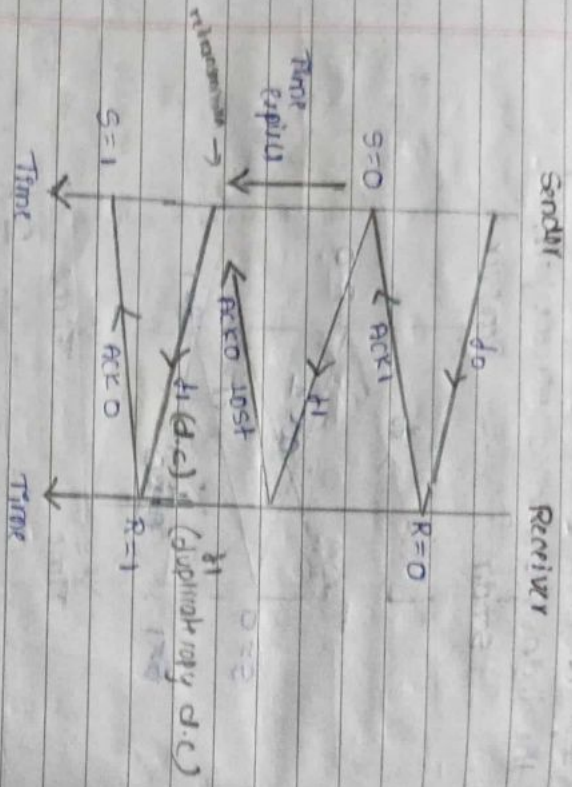
2. When ACK is lost

But due to channel condition ACK is not received.



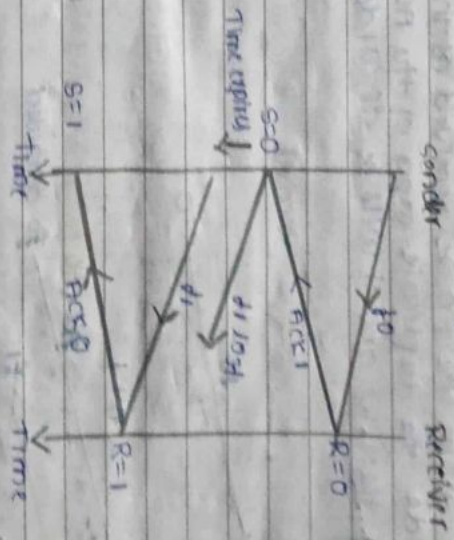
2) When ACK is lost

When the sender receives the damaged or corrupted ACK for the frame (H) it will discard the ACK and as the time expires for the ACK it will retransmit the frame (H) which will be a duplicate copy of the H. Now the receiver ~~has~~ received the correct copy of the frame and the ACK is sent at the senders end.



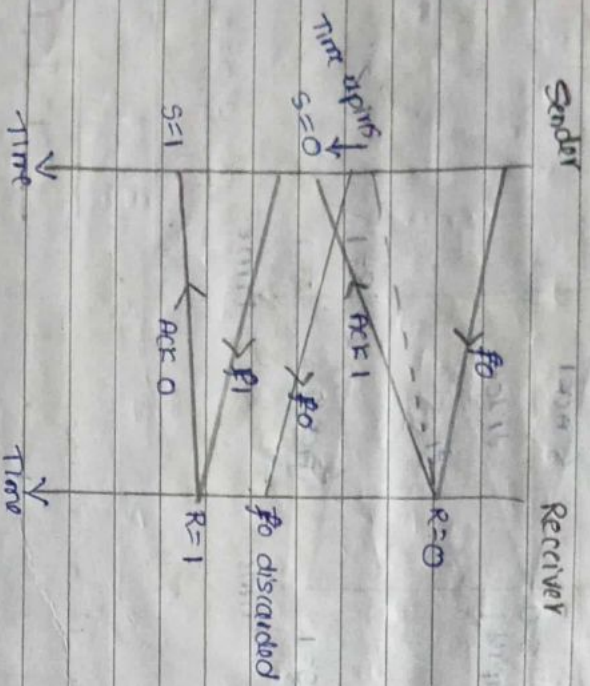
3) When Frame is lost

1) If the receiver receives corrupted or damaged frame it will discard corrupted frame and the sender will not receive ACK0 as the frame is not received by the receiver. The sender will be waiting for ACK0 till the time expires in the system and the sender will retransmit the frame (H) again and the receiver will send an ACK0 for the reply for the sent frame (H).



data medium

then when the receiver Rs not able to send the ACK-1 for the received frame (fo) in time due to the problem in ~~the~~ ^{at the} receiver end, the sender will be retransmitting fo as the ACK-1 Rs not received by the Sender in time. At the receiver end the receiver will be discarding the frame fo (duplicate) as it has already received fo but it sends the ACK-1 once again when the FO is received as the Sender side the duplicate copy of the ACK-1 ^{and} Rs already received and it will be discarded.



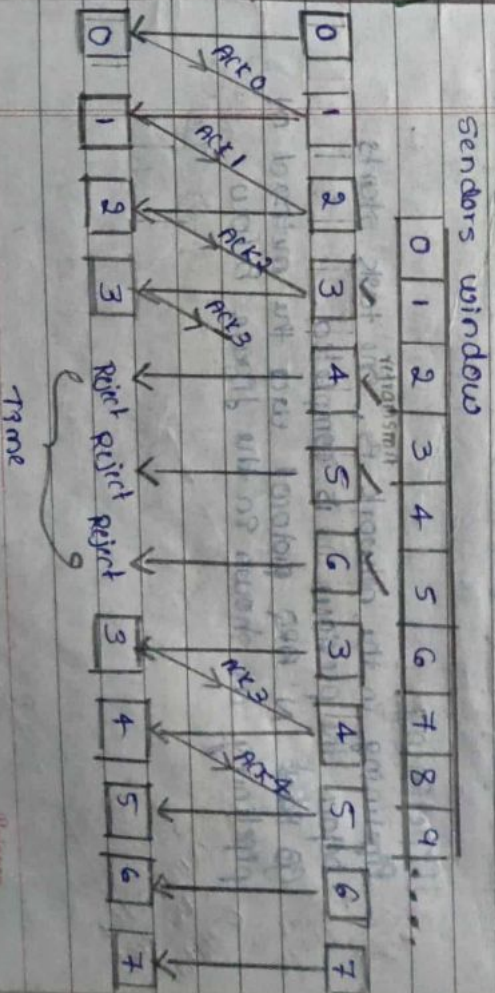
* 1 strand continuously to 3' end
 other 1 has 10% + then lost to 3' end
 retransmit 3, 4, 5, 6 retransmit
 when 3 are lost
 retransmit 6

II, 60 BACK, N1 PRQ

In this method many frames are transmitted during the period without waiting for senders ACK simultaneously the sender will have a copy of each frame sent till an ACK is received.

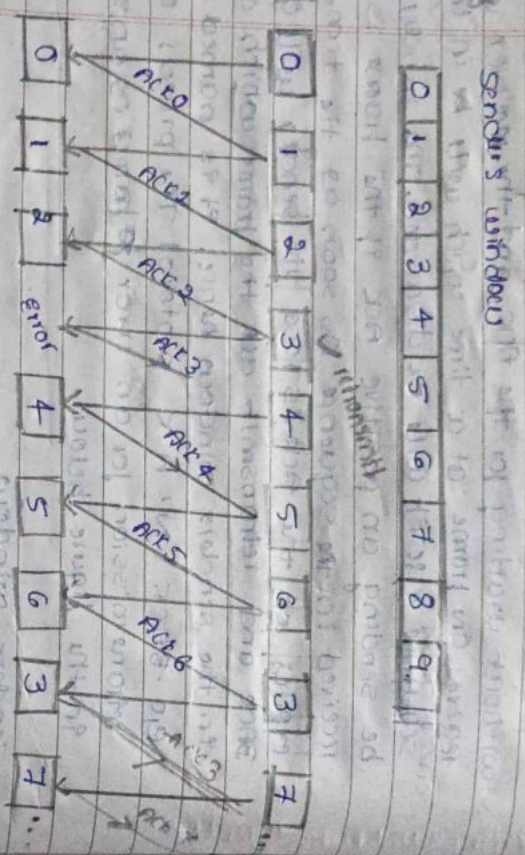
If the frame can have N sequence numbers
than the numbers will be in the range of 0 to $2^N - 1$
for example if $N = 2$ then the numbers will be
0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3

The sender can send 'N' frames continuously without waiting for the ACK and the receiver will receive one frame at a time which will be in the correct order. Else it will be discarded. The sender will sequence an positive ACK if the frame is received in the sequence as soon as the time expires on the ACK it lose the sender will go-back and retransmit all the frame which are in the senders window hence, it is named as go-back 'N' ACK protocol. The priority of retransmission for an error frame is as soon as the figure below



III Selective Repeat ARQ

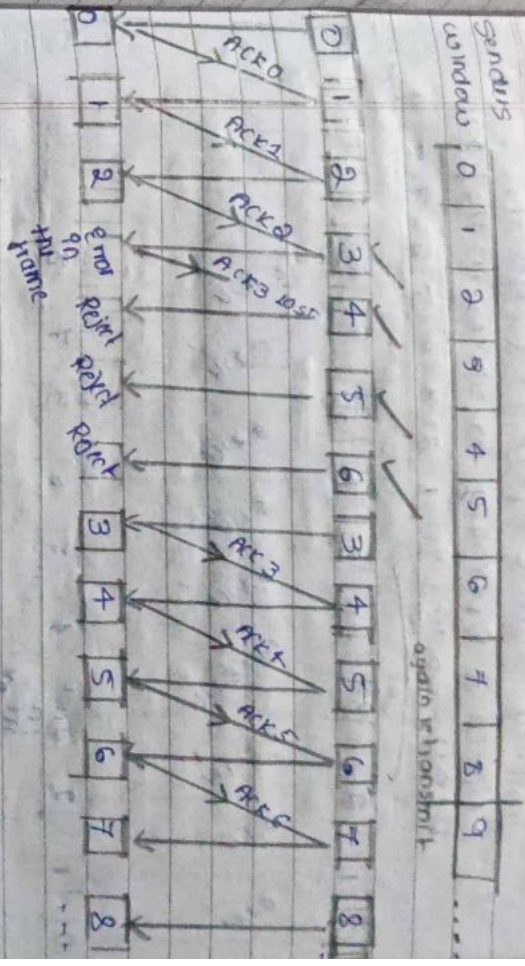
In this method the sender will retransmit which has error or not all the frames as in go-back N ARQ. Here it is named as selective repeat ARQ protocol if the ACK is not received during transmission then after the time expire the sender will retransmit only that frame which has error but not all the frames pieces of retransmission for an error frame is as seen in the figure below



IV Pipelining

Pipelining in the network is, one task starts before the previous one is completed. In go-back N ARQ protocol uses the method of pipelining as shown in the figure below

when a stop is noted on pipelining both diagram only with go-back N and selective repeat ARQ



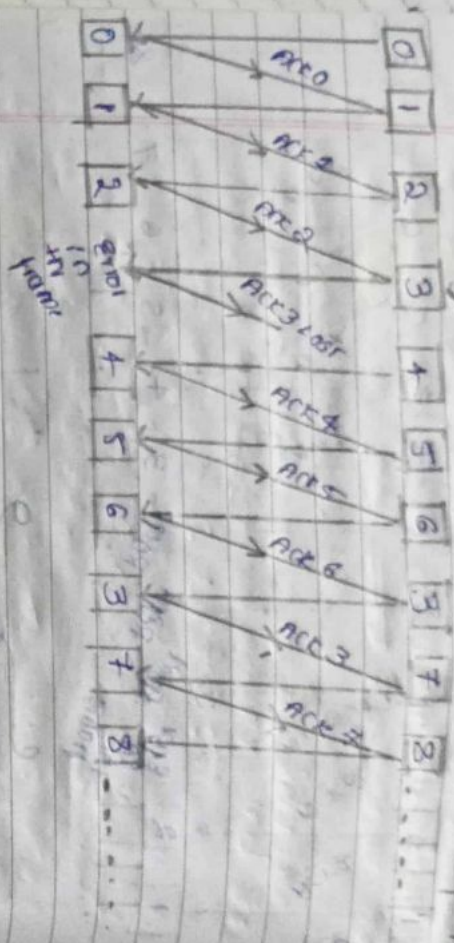
(zone) Pipelining in go-back N ARQ protocol

Stop and wait ARQ protocol does not use the method of pipelining as the sender will wait for the ACK before the next frame is sent but the method of pipelining is used in go-back N and selective repeat ARQ as in the both the methods multiple frames are sent without waiting for the ACK for the previous sent frame.

In the above figure diagram (zone) is received by the receiver without waiting for the ACK for the sent frame. This process is known as pipelining in go-back N ARQ protocol.

(b) Pipelining in Selective Repeat ARQ Protocol

Similarly Selective Repeat ARQ uses the method of pipelining as shown in the figure below.



Here frame 1 is sent received by the receiver without waiting for the ACK of the frame 0 sent by the sender's side. This process is known as pipelining in Selective Repeat ARQ protocol.

Piggy Backing

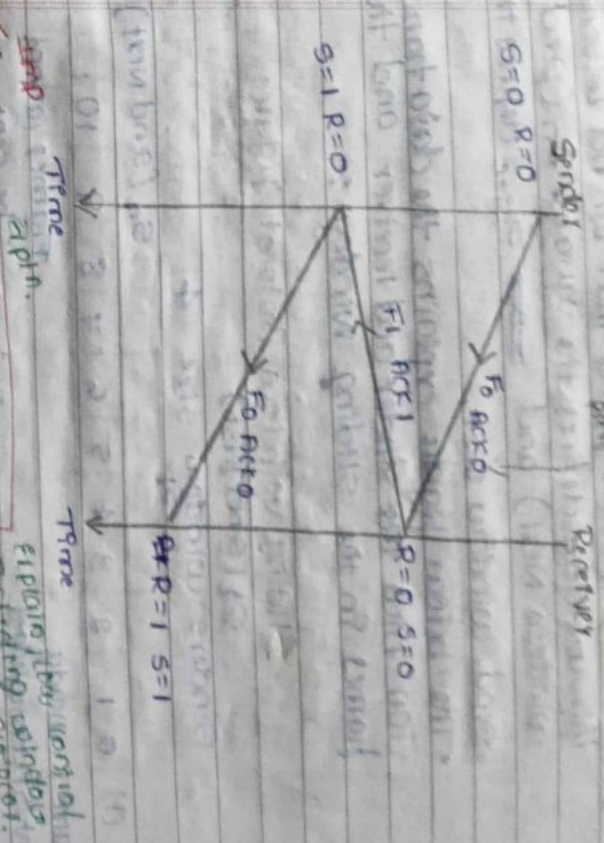
Piggy Backing is a process where it appends the ACK of the frame along with the data frame.

Piggy Backing is a process that can be used when the sender and the receiver both have some data to transmit.

- Both the sender & receiver maintain the control variable 'S' (sender) and 'R' (Receiver).
- The sender sends the frame 0 (F0) along with the appended ACK similarly the receiver will be sending

By combining both of these processes of combining piggybacking and a data frame and sending them together.

using Piggy Backing increases the overall efficiency of data transmission and the process is explained as shown in the figure below:



Sliding Window Protocol

Sliding window protocol is a concept that defines the range of sequence numbers of the sender and receiver.

The sender's range is called as 'Send sliding window' and the receiver's range is called as 'Receive sliding window'.

The maximum window size is defined as $2^N - 1$ to $2^N - 1$.

- If $N = 8$ then the window size is 4.
- The window defines the possible sequence numbers.
- The list regions are frames that are already sent, and ACK (acknowledgments).
- The second region defines the range of sequence numbers of the frame that are already sent and

This window defines the variable S_r (Send window first)

The 3rd region defines the range of sequence numbers for the frames that can not be sent this window defines the variable S_n (Send window next) and S_{size} size defines the send window size.

The below figure explains the data transmission plus the sender and receiver and the frames in the sliding window:

Sliding window Protocol figure:

S_1 (send first)

S_n (send next)

Senders window size 4

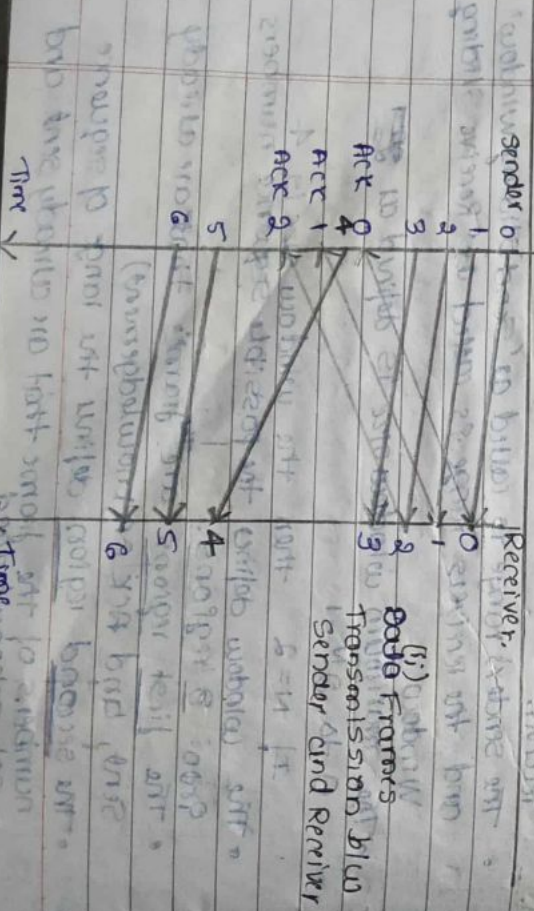
0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

frames already sent & acknowledged

frames sent but not acknowledged

frames cannot be sent

Send sliding window



Types of Error

There are 3 types of the errors that may occur during the data transmission over a network.

Types

- 1-bit error $\rightarrow$ only 1 bit is changed
- Burst error $\rightarrow$ 2 or more bits are changed.
- len of msg/frame $\rightarrow$ during transmission data not sent properly.

① 1-bit error

1-bit error or single bit error is caused when only one bit is changed in the data during the transmission from the sender to the receiver. That is 1 is changed to 0 or 0 is changed to 1.

0	1	0	1	0	0	1	0
---	---	---	---	---	---	---	---

(Sent by sender)

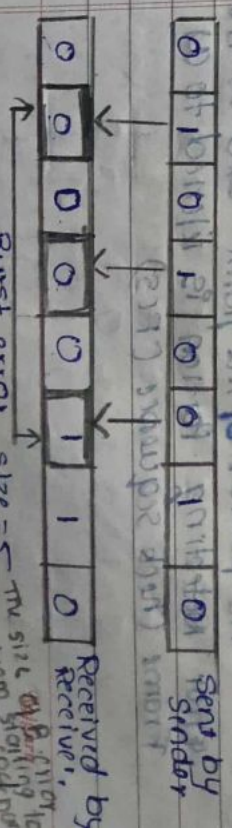
0	1	0	0	0	0	1	0
---	---	---	---	---	---	---	---

(Received by Receiver)

② Burst error

Burst error means that 2 or more bits of the data is changed during transmission from sender to receiver.

The size of burst error is from the first corrupted bit to the last corrupted bit.



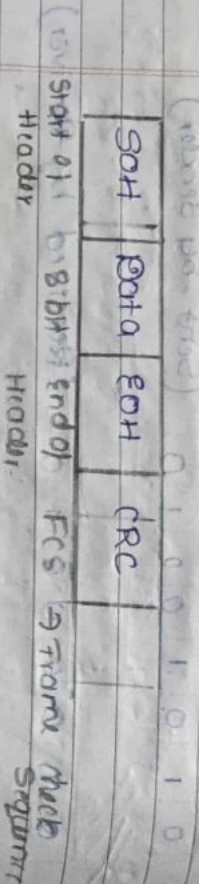
Types of Error $\rightarrow$ 2M
Reason Error $\rightarrow$ 2M
Frame - 3 bit

3) Loss of msg) Frame

- The sender has sent the frame but it is not received correctly by the receiver. This is known as loss of frame during Transmission.
- To deal with this type of error transmission of the frame is required by the sender.

Format of Frame

- Most of the Error Detecting schemes attach the Error Detection Information to the data frame before it is sent by the sender.
- The transmitted frame is shown in the figure below which includes Error Detection Information



- The frame consists of following fields other than the data field

- SOH → Indicate start of the header which is the beginning of the data frame.
- EOH → Indicate the end of header which is the ending of data transmission.
- FCS → The portion of the frame used as the error detecting portion is referred to as Frame Check Sequence (FCS)

10.4

explains error detection - 10.4
error correction - 10.5

10.4 Error Detection Methods

Error detecting methods or Redundancy check methods commonly used in data transmission are:

- 1) Parity Bit Check Method
- 2) Check Sum Method
- 3) CRC - Cyclic Redundancy Check Method

1) Parity bit Check Method: Parity check is a mechanism which is used to verify whether the transmitted data has been corrupted during communication at the receiver's end.

In this method a 1 extra bit called Parity bit is added to the data before transmission.

- This parity bit is set by the transmitter to the group of bits and when the receiver receives the data the parity bit will be removed and recomputed with the group of bits and verified with the Parity bit set.

There are 2 types of Parity Check Methods

- 1) Even Parity Bit
- 2) Odd Parity Bit

1) Even Parity Bit: The total number of 1's (including parity bit) in a block of data with the even number of 1's.

Consider the example below, the 9th column contains parity bit.

1	0	1	0	0	1	0	1	0	1
1	0	1	0	1	0	0	1	0	1
1	0	1	0	1	0	0	1	0	1
1	0	1	0	1	0	0	1	0	1

100100100
100100100

00000000; 00000000

	A	B	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	0	1	1	0
1	0	1	1	0
1	1	0	0	1
1+1+1				

$$1+1+1+1+1=5$$

0	1	2	3
0	1	2	3

100	1	0	0	0	1	101
only sum	1	0	0	1	010	sum

01101101

[illegible]

011011011:000100001

◦ In this method the checksum generator divides the given input data into equal segments of k^{th} and the addition of these bits is carried out and the obtained the one's complement to the sum. This result is known as Checksum.

- At the receiver, we need to add all the bits and if the result is all 1's (ones) then the data will be received as Error Free and the complement of the same is the all zeros. Otherwise it will be rejected.

Casing Error Detection Method - Checksum

11010101

Gold 40; 100 11 00 11

$$\boxed{10101101}$$

0.01111

101101

10000110

1 1 1 0 0 0 1 1 1

15 Complement of 01110000 [Checksum]

Data with appended checksum is 01

01100000

Receiver side

Data 1: 1 1 0 0 1 1 0 1 1

Data 2: 1 0 1 0 1 0 1 1

1 0 1 0 1 0 1 1

0

Data 3: 0 1 1 0 1 1 0 1 0

1 1 0 1 1 1 0 1 0

Data 4: 1 1 0 1 0 1 0 1 0 1

1 1 0 1 0 1 1 1 0

1 1 0 1 0 1 1 1 0

Data 5: 1 1 0 1 0 1 1 1 1

1 1 0 1 0 1 1 1 1

1's complement 0 0 0 0 0 0 0 0 [checksum]

As the complement is 0 (zero) it indirectly that the data is error free which is received at the receiver end so

it will be accepted by the receiver.

2) Consider the data unit that is to be transmitted, 10010010 - 0000111100 11001011

Sender's data is 01

10010010 00001111 11001100

10110110

Sender's side

Data 1: 1 0 0 1 0 0 1 0

Data 2: 0 0 0 0 1 1 1 1

1 0 1 0 0 0 0 1

Data 3: 1 1 1 0 0 1 1 0 0

1 1 0 1 1 0 1 1 0 1

1 0 1 1 0 1 1 1 0

Data 4: 1 0 1 1 0 1 1 0 1 1 0

1 1 1 0 1 0 1 0 1 0 0

Data 5: 1 0 0 0 0 0 1 0 1 0 1

1's complement 1 1 0 1 1 0 1 0 [checksum]

Data with appended checksum is 01

0110010010 00001111 11001100

11001100 10110110 11011011

Receiver's Side

Data:

Receiver's Side

Data 1: 1 0 0 1 0 0 1 0

Data 2: 0 0 0 0 1 1 1 1

Data 3: 1 0 1 0 0 0 0 1

Data 4: 1 1 0 0 1 1 0 0

Data 5: 1 1 0 1 1 0 1 0

Data 6: 1 0 1 1 0 1 1 0

Data 7: 1 0 1 1 0 1 1 0

Data 8: 1 0 1 1 0 1 1 0

Data 9: 1 0 1 1 0 1 1 0

Data 10: 1 0 1 1 0 1 1 0

Data 11: 1 0 1 1 0 1 1 0

Data 12: 1 0 1 1 0 1 1 0

Data 13: 1 0 1 1 0 1 1 0

Data 14: 1 0 1 1 0 1 1 0

Data 15: 1 0 1 1 0 1 1 0

Data 16: 1 0 1 1 0 1 1 0

Data 17: 1 0 1 1 0 1 1 0

Data 18: 1 0 1 1 0 1 1 0

Data 19: 1 0 1 1 0 1 1 0

Data 20: 1 0 1 1 0 1 1 0

Data 21: 1 0 1 1 0 1 1 0

Data 22: 1 0 1 1 0 1 1 0

Data 23: 1 0 1 1 0 1 1 0

Data 24: 1 0 1 1 0 1 1 0

Data 25: 1 0 1 1 0 1 1 0

Data 26: 1 0 1 1 0 1 1 0

Explain polynomial code with example.

3) Cyclic Redundancy Check (CRC) / Polynomial Code

The sender & the receiver should agree to share the generator polynomial code before the transmission process starts.

CRC generation at the sender's side:

1) Find the length of the divisor '1'

2) Append 'L-1' bits to the original message

3) Perform binary division operation

4) Remainder of the division is CRC

5) Append the remainder with the original message and transmit the data to the receiver

Note:

The CRC must be of 'L-1' bits

Now append the remainder with the original message and transmit the data to the receiver

After CRC generation detecting error in the receiver's side:

1) Divide the received bits by using the same generator polynomial

2) If the remainder is all 0's then there is no error in the transmission data sent by the sender

3) Determine the transmitted CRC for following data, 10 and the generator code is verify the remainder is zero

given that, 10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

10010010000001

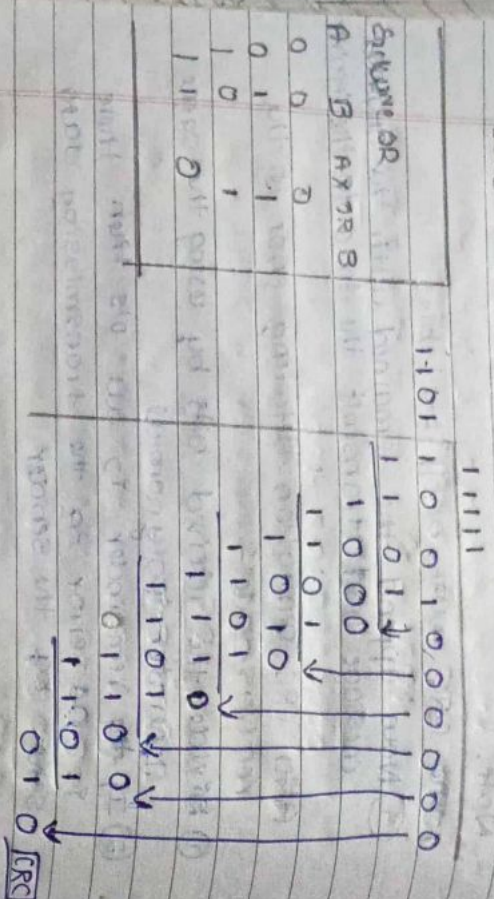
10010010000001

Soln: Since the generator code has 4 data bits, add 3 zero bits ($L-1 = 4-1 = 3$) to the data the appended data is

0111000100000000

Now, divide the appended data by the generator code using binary division operation until all the bits are used

Sender's Side



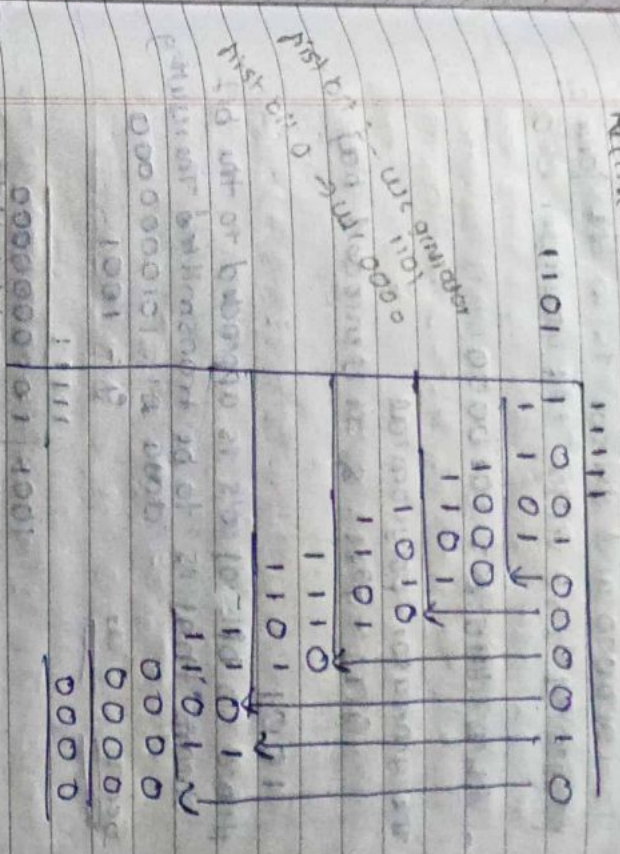
Now append this remainder (CRC) along with the original data.

1001000000
010 [CRC]
10010000010

This data is transmitted to the receiver

The data is transmitted to the Receiver

1101



When the receiver gets all 0's has the remainder the receiver understand that there are no transmission errors in the data sent by the sender.

Perform the checksum

1001000000001111001100101010110

110
0000000001
110
11000000101

110
11000000101

3) Consider the message sender wants to send is 1010001101 and the generator polynomial is $x^5 + x^4 + x^2 + 1$. Find the message that is to be transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.

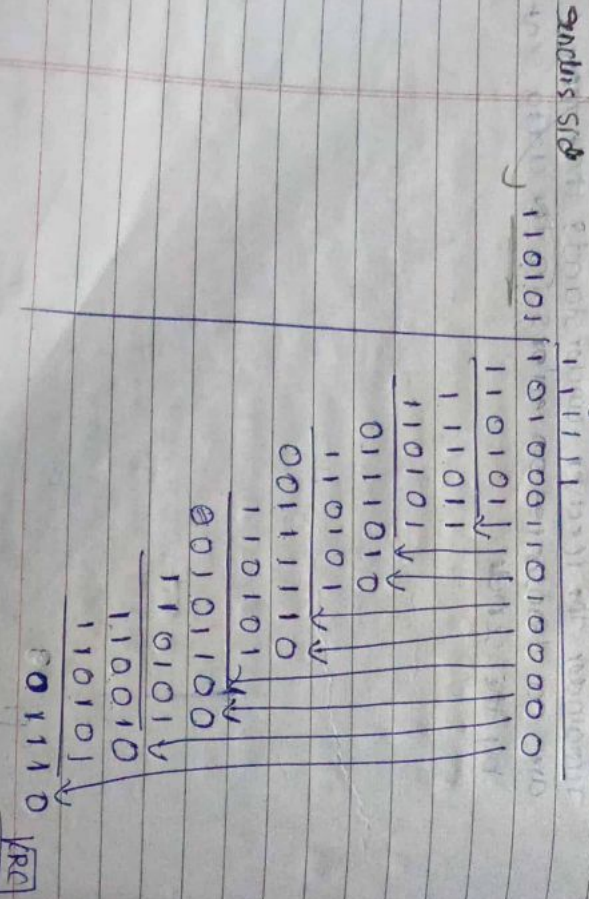
Soln: Given Data: $D = 1010001101$

The generator polynomial $g(x) = x^5 + x^4 + x^2 + 1$
 & it is encoded has 110101

x^5 x^4 x^3 x^2 x^1 x^0
 1 0 1 0 1 0 1

hence 5 bits of 0's is appended to the bit stream that is to be transmitted. The resulting data $D' = 1010001100000$

Sender side

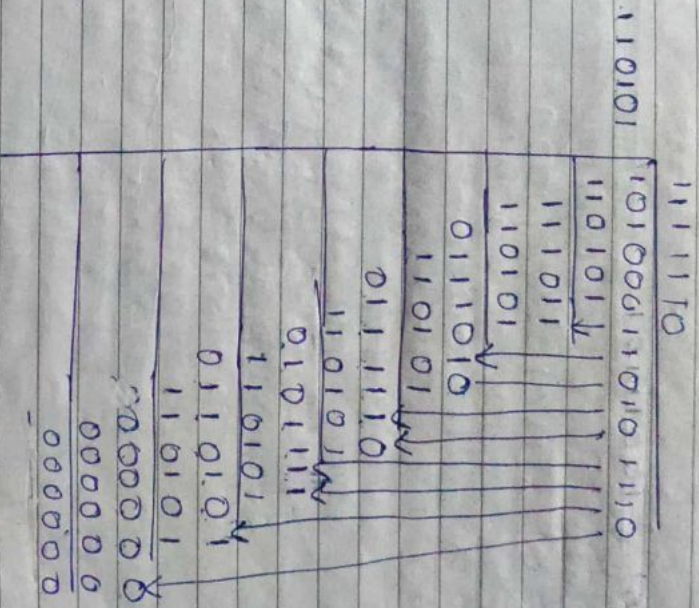


Now append the remainder (RC) with the original data

$\begin{array}{r} 101000110100000 \\ + \quad 01110 \\ \hline 101000110101110 \end{array}$

The data is transmitted to the receiver.

Receiver side



When the receiver gets all 0's has the remainder the receiver understands that there are no transmission errors in the data sent by the sender.

4) A bit stream 10011001 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$.

Suppose the 3rd bit from the left is inverted or changed during transmission. How will the receiver detect this error?

$x^3 \quad x^2 \quad x^1 \quad x^0$
 0 0 0
 0 1 1
 1 0 1

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

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$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

$$D = 10011001 + 1001 = 10101100$$

1001100100

1001100100

Here the CRC is 100. Therefore the code word transmitted to the receiver is 1001100100.

According to the given question the 3rd bit from the left gets changed during transmission. Now the transmitted bits are 1011100100.

Receiver's side

101010
 1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

1001101100100

The remainder obtained on division is a non-zero value. This indicates to the receiver that an error has occurred in the data during transmission. Therefore the receiver rejects the data and requests the sender for retransmission.

② Given data $D = 1101011011$ and generator code $g = 10011$

$$L = 5$$

$$s-1 = 4$$

$$g = 10011$$

$$D' = 110101101100001$$

$$1100010100010$$

$$100110101100000$$

$$0110011$$

$$10011$$

$$000001$$

$$00000$$

$$00010$$

$$00101$$

$$00000$$

$$00101$$

$$00000$$

$$00101$$

$$010110$$

$$10011$$

$$001010$$

$$000000$$

$$01010$$

$$00000$$

$$01010$$

$$01010$$

$$1001$$

$$001110$$

$$00000$$

$$1110$$

[CRC]

the CRC is 1110. Therefore the word transmitted to the receiver is

$$1101011011110$$

Now append this remainder (CRC) along with the original data

$$11010110110000$$

$$1110$$

$$1101011011110$$

This data is transmitted to the receiver.

Receiver's side

$$1101011011110$$

$$100110101101110$$

$$10011$$

$$10011$$

$$000001$$

$$0000010$$